

Deep Learning for Data Science

DS 542

<https://dl4ds.github.io/fa2025/>

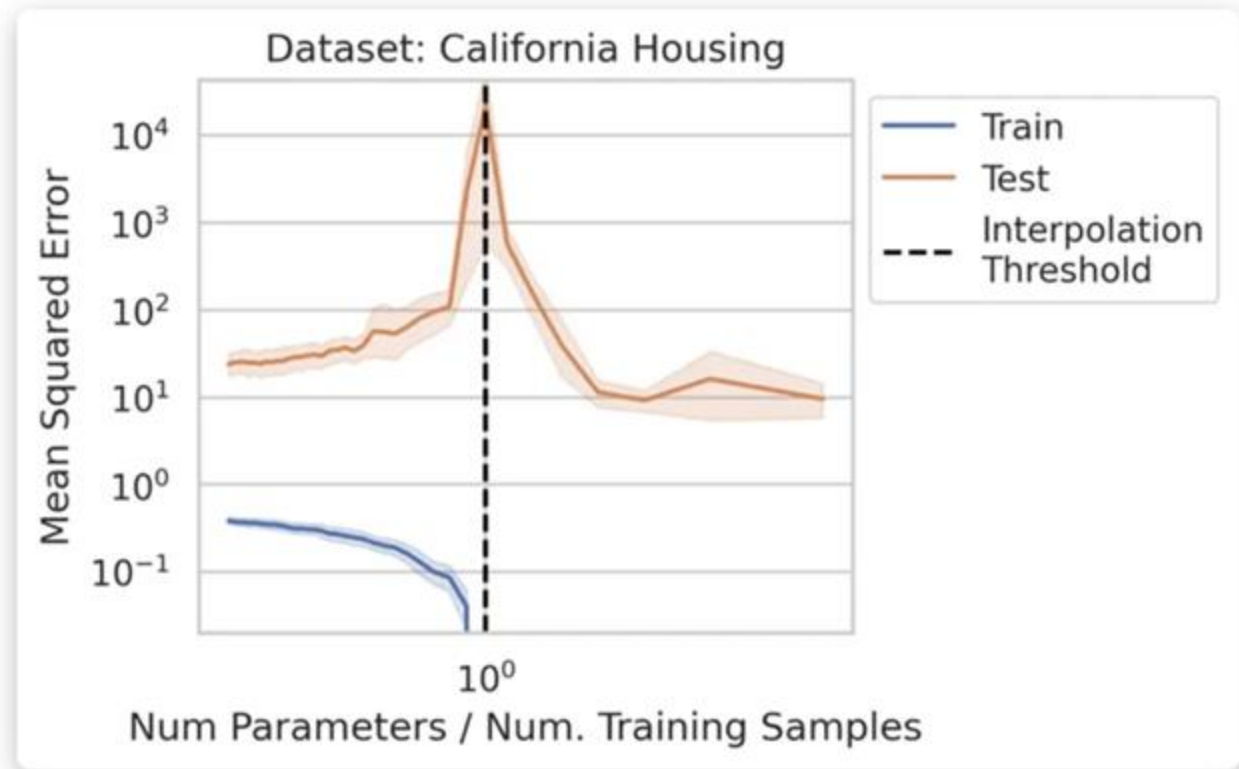
Regularization

Plan for Today

- Double descent
- Theoretical results on generalization (brief)
- Regularization

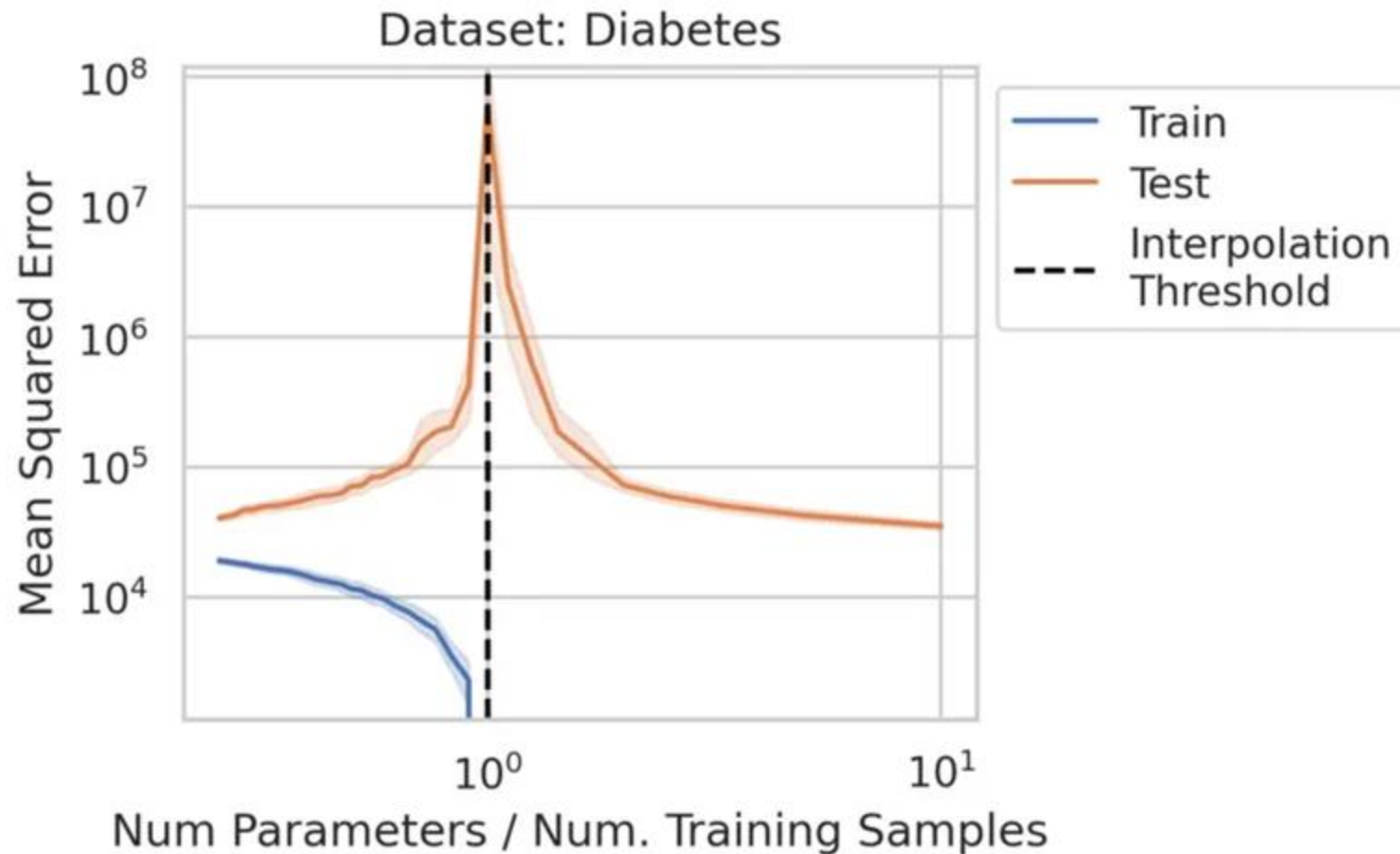
Double Descent Demystified

- Linear regression too?



<https://iclr-blogposts.github.io/2024/blog/double-descent-demystified/>

Reproduced with Classic Data Sets



Double Descent Test Setup

- Used polynomial features to generate arbitrary numbers of features
- When more polynomial features than training samples, regression has multiple parameters for exact fit.
 - Pick parameters minimizing a norm $\leftarrow$ this is a regularization.

Rough Explanation for Double Descent

1. Parameters $\ll$ samples

- Model can only fit overall trends. Cannot fit individual points particularly well.
- Training and test loss improve with more parameters.

2. Parameters $\sim$ samples

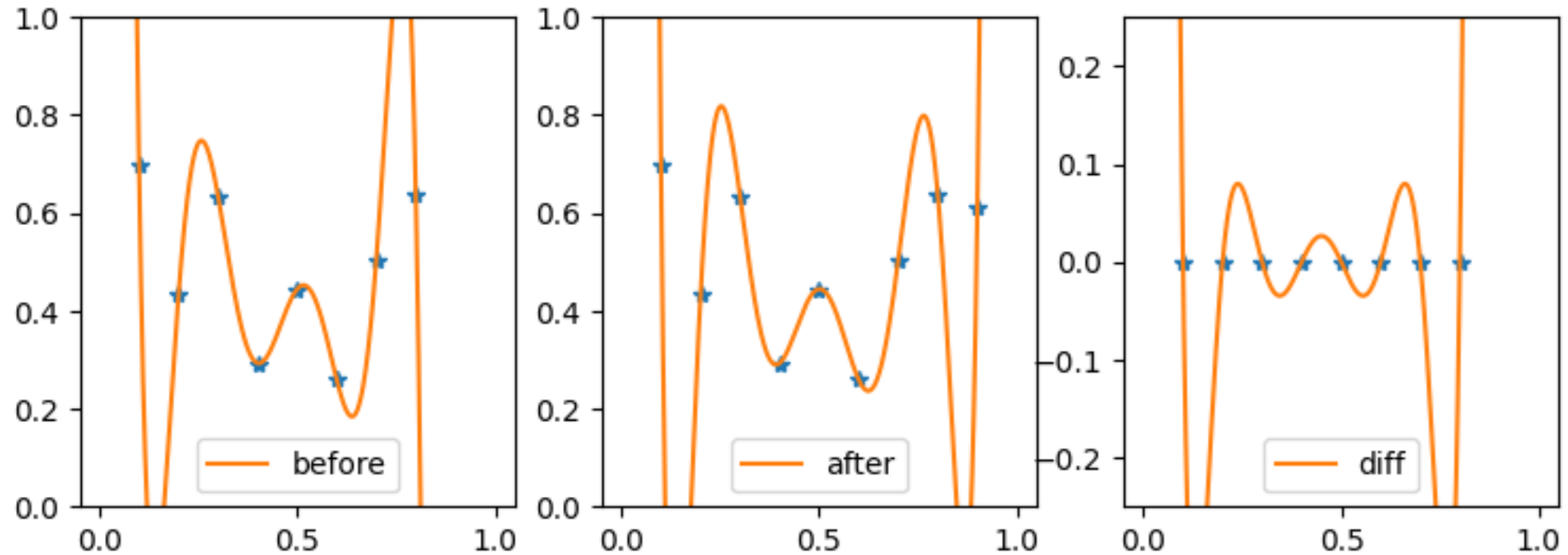
- Model can barely / not quite fit all training points.
- Contortions are likely.
- Detailed analysis says “singular values” a lot. Read the paper if curious.

3. Parameters $\gg$ samples

- Model can easily fit all training points.
- Lots of freedom to make parameter norms smaller.
- Some intuitive and proven connections to better generalization from smaller norms.

Re: Contortions are Likely

Original slide title: What Happens When You Add Another Point?



Any Questions?

???

Theoretical Results about Generalization

- TODO like sum of weights

Theoretical Results

- For Valid Generalization the Size of the Weights is More Important than the Size of the Network
- Train faster, generalize better: Stability of stochastic gradient descent

Watch for these themes as we go through regularization examples.

Any Questions?

???

Regularization

- Why is there a generalization gap between training and test data?
 - Overfitting (model describes statistical peculiarities)
 - Model unconstrained in areas where there are no training examples
- **Regularization** = methods to reduce the generalization gap
- Technically means adding terms to loss function
- But colloquially means any method (hack) to reduce gap between training and test data

Regularization

- Explicit regularization
- Implicit regularization
- Early stopping
- Ensembling
- Dropout
- Adding noise
- Transfer learning, multi-task learning, self-supervised learning
- Data augmentation

Explicit regularization

- Standard loss function:

$$\begin{aligned}\hat{\phi} &= \operatorname{argmin}_{\phi} [L[\phi]] \\ &= \operatorname{argmin}_{\phi} \left[\sum_{i=1}^I \ell_i[\mathbf{x}_i, \mathbf{y}_i] \right]\end{aligned}$$

Explicit regularization

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- Regularization adds an extra term

$$\hat{\phi} = \operatorname{argmin}_{\phi} \left[\sum_{i=1}^I \ell_i[\mathbf{x}_i, \mathbf{y}_i] + \lambda \cdot g[\phi] \right]$$

Explicit regularization

- Standard loss function:

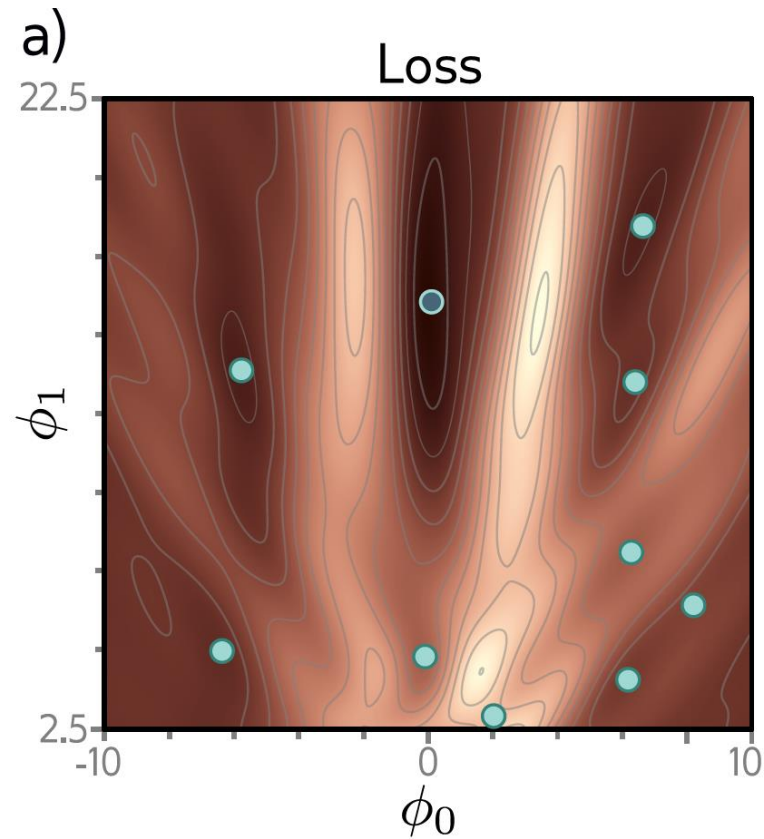
$$\begin{aligned}\hat{\phi} &= \operatorname{argmin}_{\phi} [L[\phi]] \\ &= \operatorname{argmin}_{\phi} \left[\sum_{i=1}^I \ell_i[\mathbf{x}_i, \mathbf{y}_i] \right]\end{aligned}$$

- Regularization adds an extra term

$$\hat{\phi} = \operatorname{argmin}_{\phi} \left[\sum_{i=1}^I \ell_i[\mathbf{x}_i, \mathbf{y}_i] + \lambda \cdot g[\phi] \right]$$

- Where $g[\phi]$ is smaller for preferred parameters
- $\lambda > 0$ controls the strength of influence

Explicit regularization

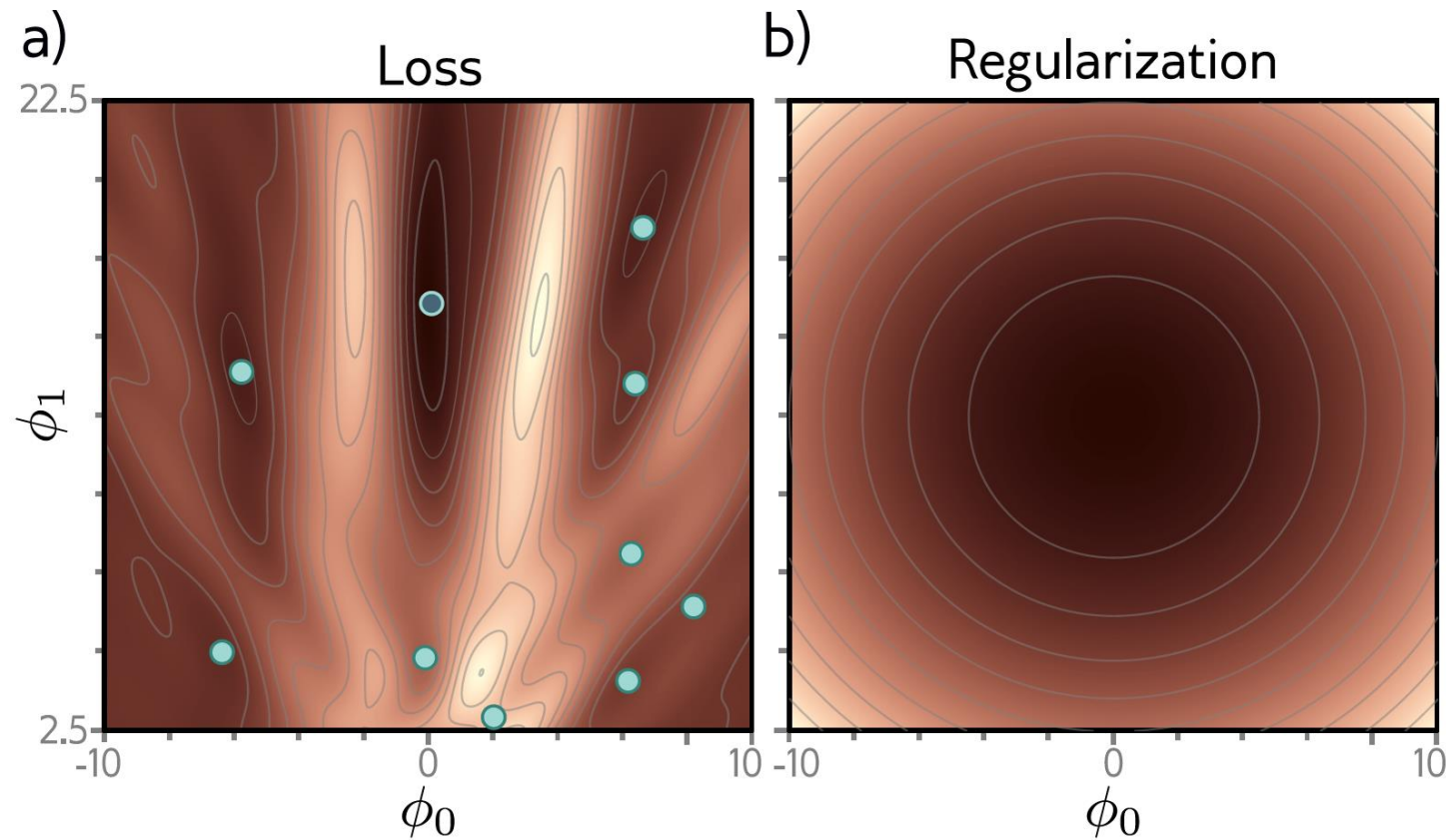


Loss function for Gabor model of Lecture 6 and Chapter 6.



denotes local minima

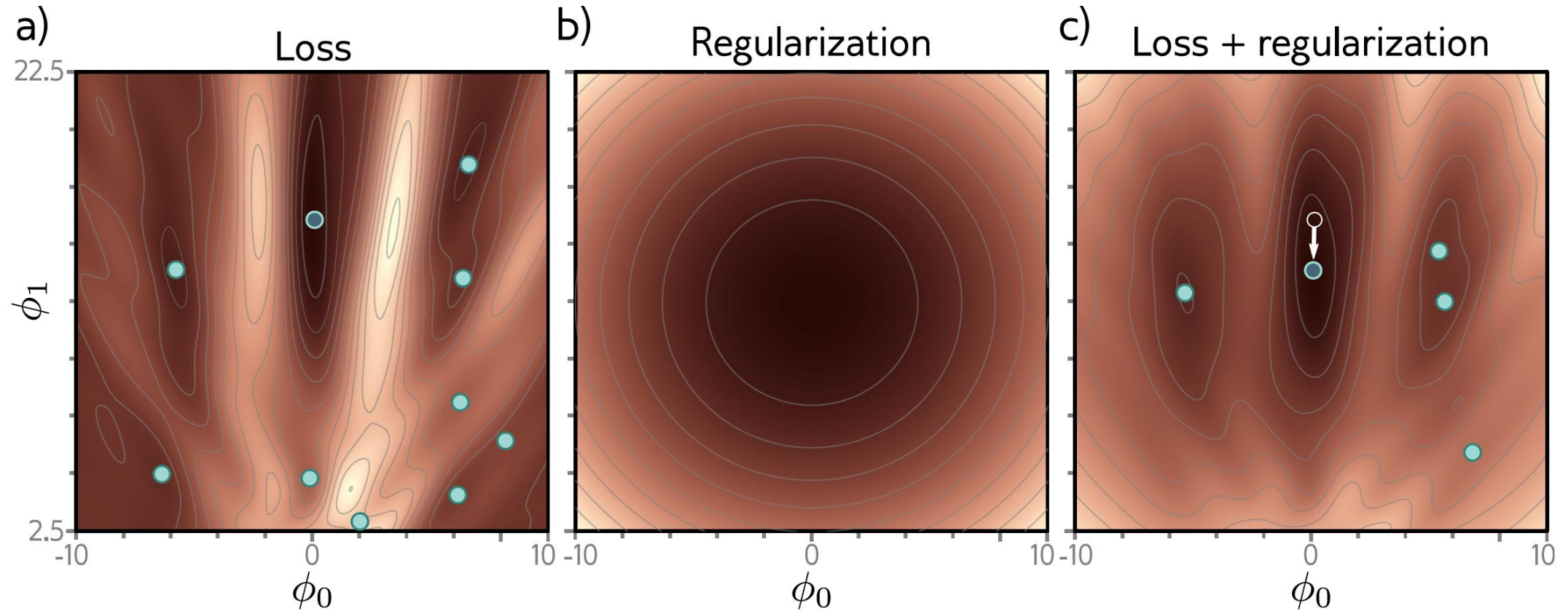
Explicit regularization



Example of a regularization function that prefers parameters close to 0.

Explicit regularization

Fewer local minima and the absolute minimum has moved.



● denotes local minima

Probabilistic interpretation

- Maximum likelihood:

$$\hat{\phi} = \operatorname{argmax}_{\phi} \left[\prod_{i=1}^I \operatorname{Pr}(\mathbf{y}_i | \mathbf{x}_i, \phi) \right]$$

- Regularization is equivalent to adding a **prior** over parameters

$$\hat{\phi} = \operatorname{argmax}_{\phi} \left[\prod_{i=1}^I \operatorname{Pr}(\mathbf{y}_i | \mathbf{x}_i, \phi) \operatorname{Pr}(\phi) \right] \quad \text{Maximum a posteriori or MAP criterion}$$

... what you know about parameters *before* seeing the data

Equivalence

- Explicit regularization:

$$\hat{\phi} = \operatorname{argmin}_{\phi} \left[\sum_{i=1}^I \ell_i[\mathbf{x}_i, \mathbf{y}_i] + \lambda \cdot g[\phi] \right]$$

- Probabilistic interpretation:

$$\hat{\phi} = \operatorname{argmax}_{\phi} \left[\prod_{i=1}^I Pr(\mathbf{y}_i | \mathbf{x}_i, \phi) Pr(\phi) \right]$$

- Converting to Negative Log Likelihood (e.g. $-\log(\cdot)$):

$$\lambda \cdot g[\phi] = -\log[Pr(\phi)]$$

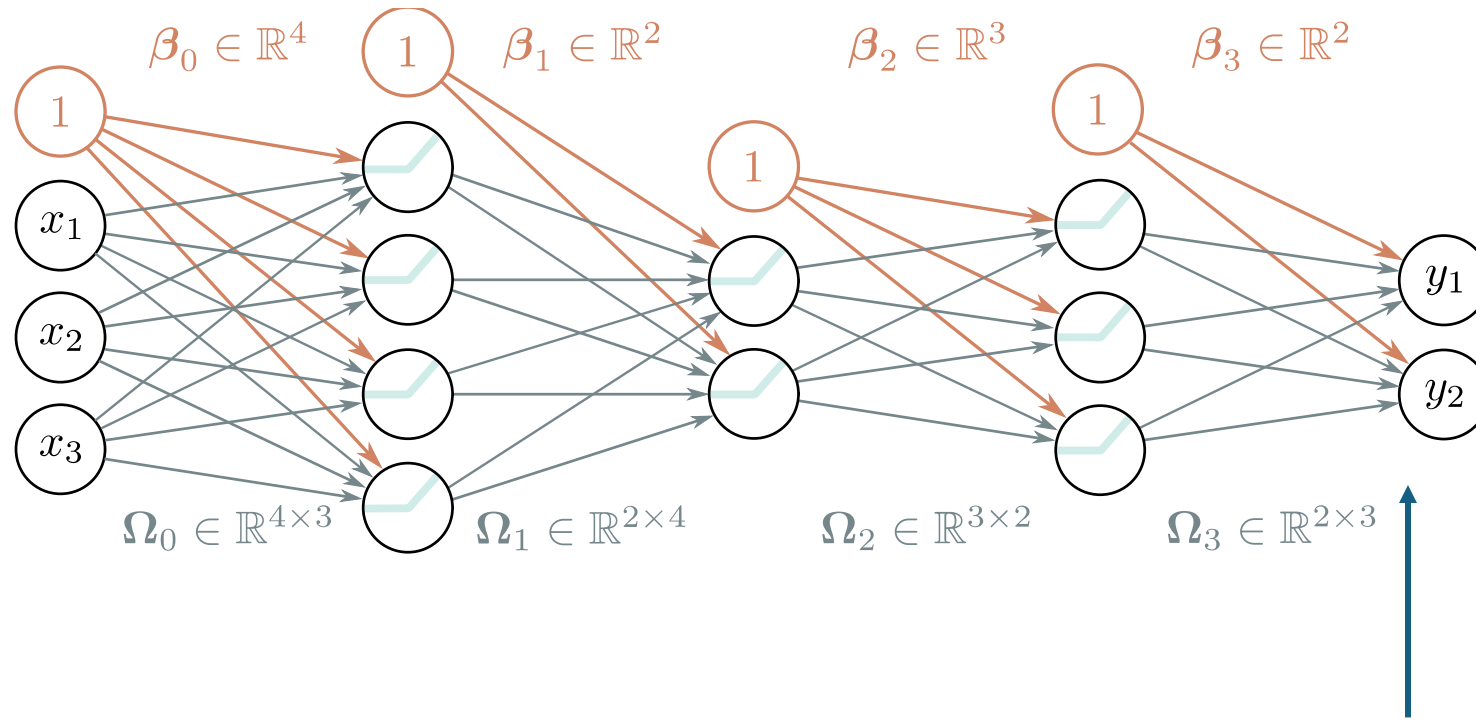
L2 Regularization

- Most common regularizer is **L2 regularization**
- Favors smaller parameters (like in previous example)

$$\hat{\phi} = \operatorname{argmin}_{\phi} \left[L[\phi, \{\mathbf{x}_i, \mathbf{y}_i\}] + \lambda \sum_j \phi_j^2 \right]$$

- Also called **Tikhonov regularization, ridge regression**
- In neural networks, usually just for weights, and called **weight decay**

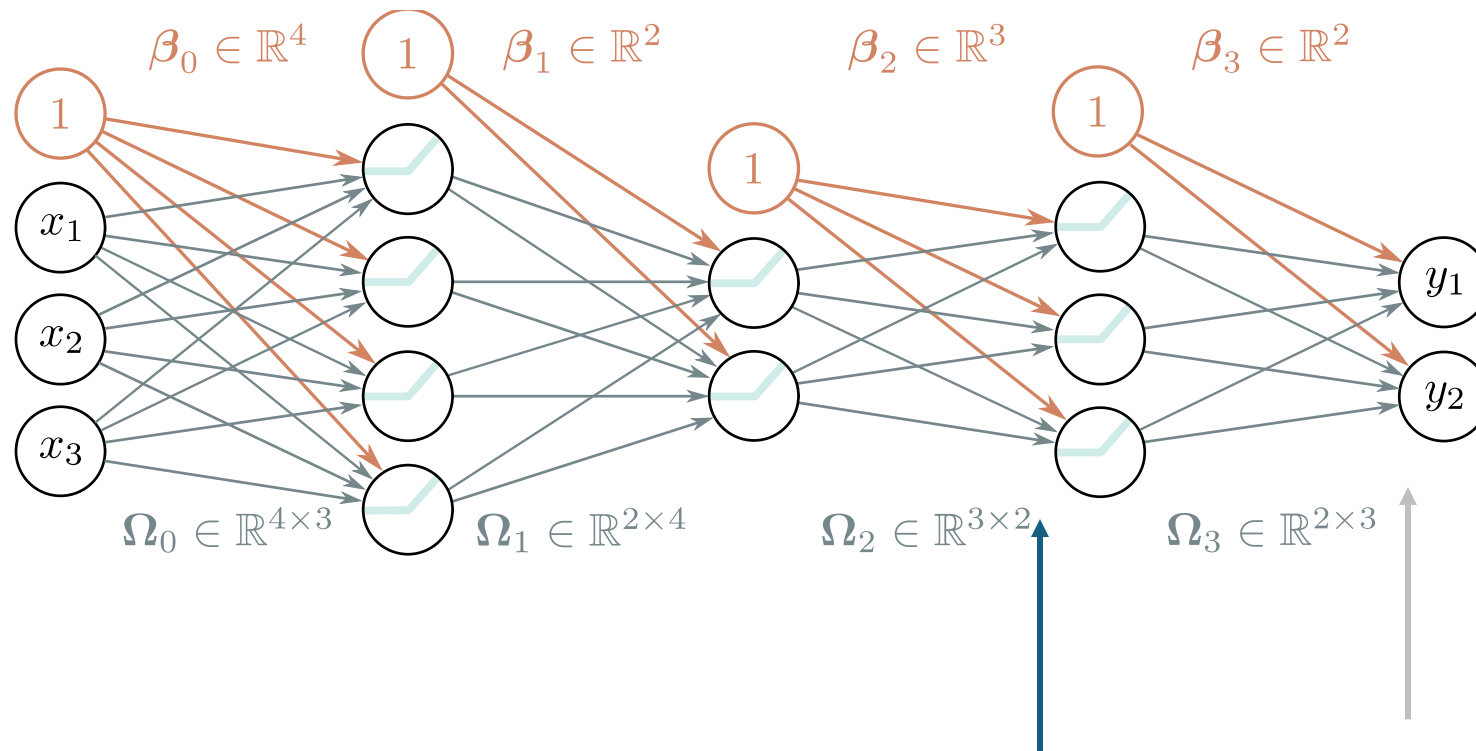
Why does L2 regularization help?



Outputs are weighted linear combination of last layer activations.

Smaller weights attenuate changes.

Why does L2 regularization help?

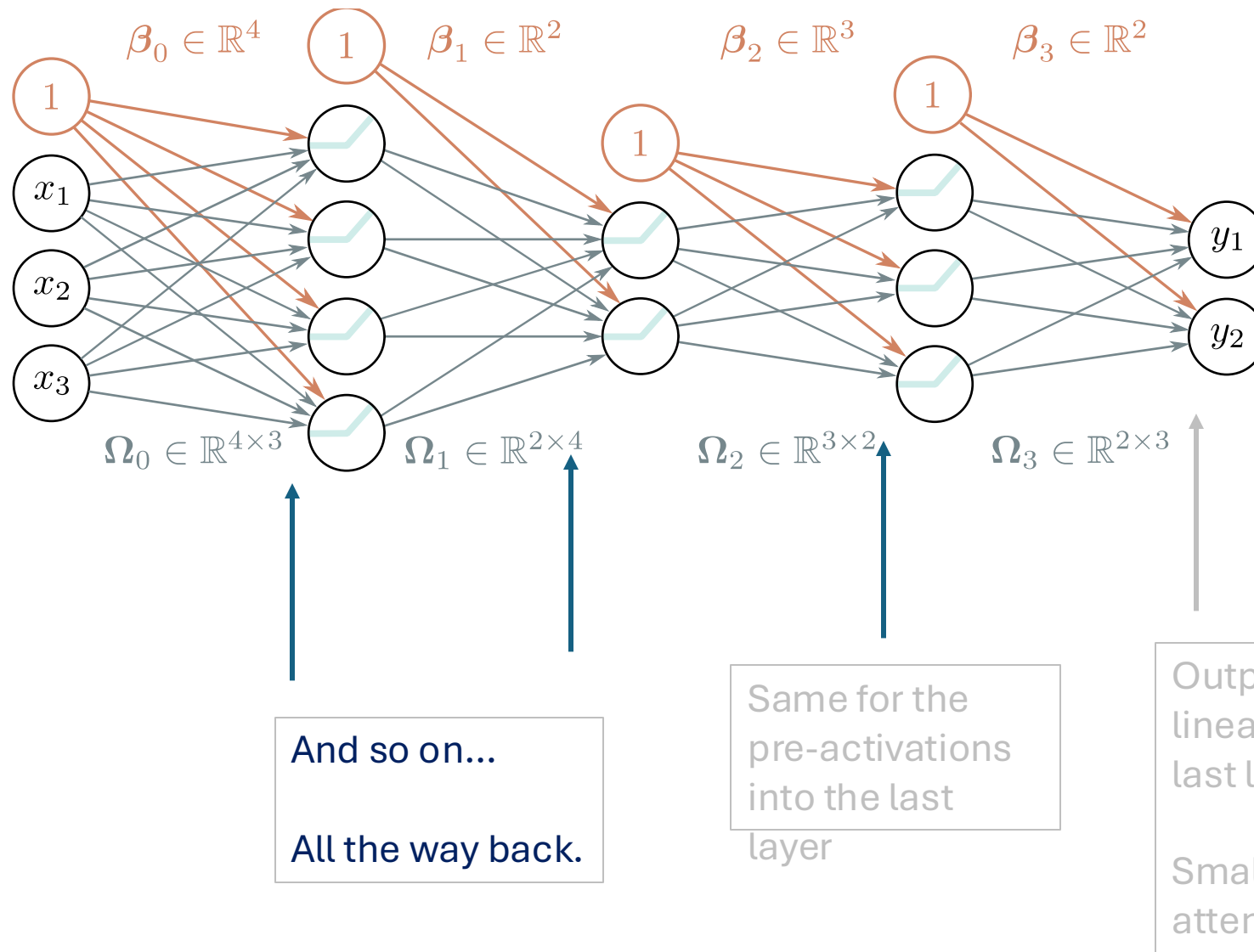


Same for the
pre-activations
into the last
layer

Outputs are weighted
linear combination of
last layer activations.

Smaller weights
attenuate changes.

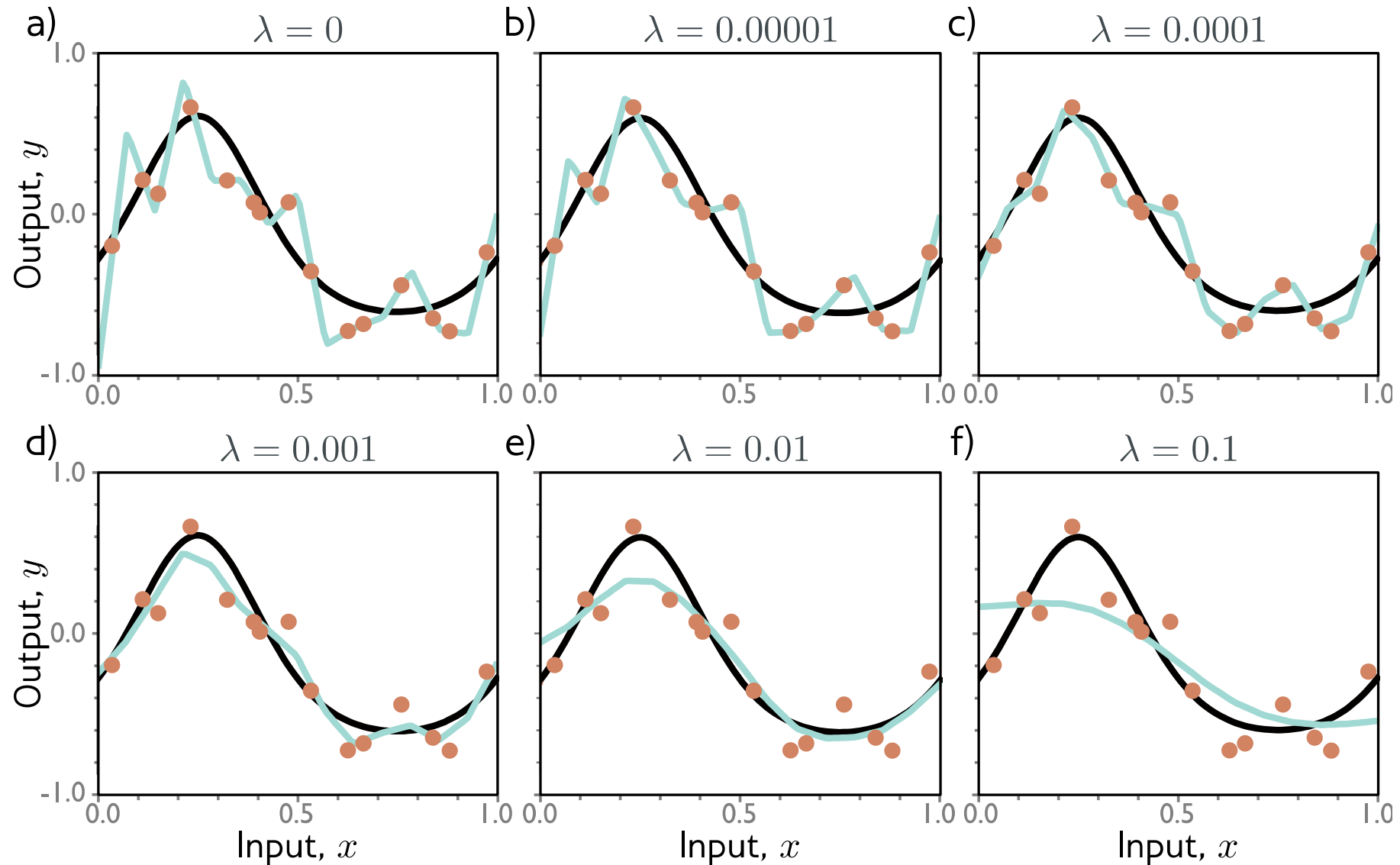
Why does L2 regularization help?



Why does L2 regularization help?

- Discourages fitting excessively to the training data (overfitting)
- Encourages smoothness between datapoints

L2 regularization (simple net from last lecture)



PyTorch Explicit L2 Regularizer

SGD

```
CLASS torch.optim.SGD(params, lr=0.001, momentum=0, dampening=0, weight_decay=0,  
                     nesterov=False, *, maximize=False, foreach=None, differentiable=False) [SOURCE]
```

Implements stochastic gradient descent (optionally with momentum).

Parameters

- **params** (*iterable*) – iterable of parameters to optimize or dicts defining parameter groups
- **lr** (*float*, *optional*) – learning rate (default: 1e-3)
- **momentum** (*float*, *optional*) – momentum factor (default: 0)
- **weight_decay** (*float*, *optional*) – weight decay (L2 penalty) (default: 0)

<https://pytorch.org/docs/stable/generated/torch.optim.SGD.html>

ADAM

```
CLASS torch.optim.Adam(params, lr=0.001, betas=(0.9, 0.999), eps=1e-08,  
                      weight_decay=0, amsgrad=False, *, foreach=None, maximize=False,  
                      capturable=False, differentiable=False, fused=None) [SOURCE]
```

Implements Adam algorithm.

Parameters

- **params** (*iterable*) – iterable of parameters to optimize or dicts defining parameter groups
- **lr** (*float*, *Tensor*, *optional*) – learning rate (default: 1e-3). A tensor LR is not yet supported for all our implementations. Please use a float LR if you are not also specifying *fused*=True or *capturable*=True.
- **betas** (*Tuple*[*float*, *float*], *optional*) – coefficients used for computing running averages of gradient and its square (default: (0.9, 0.999))
- **eps** (*float*, *optional*) – term added to the denominator to improve numerical stability (default: 1e-8)
- **weight_decay** (*float*, *optional*) – weight decay (L2 penalty) (default: 0)

<https://pytorch.org/docs/stable/generated/torch.optim.Adam.html>

Regularization

- Explicit regularization
- Implicit regularization
- Early stopping
- Ensembling
- Dropout
- Adding noise
- Transfer learning, multi-task learning, self-supervised learning
- Data augmentation

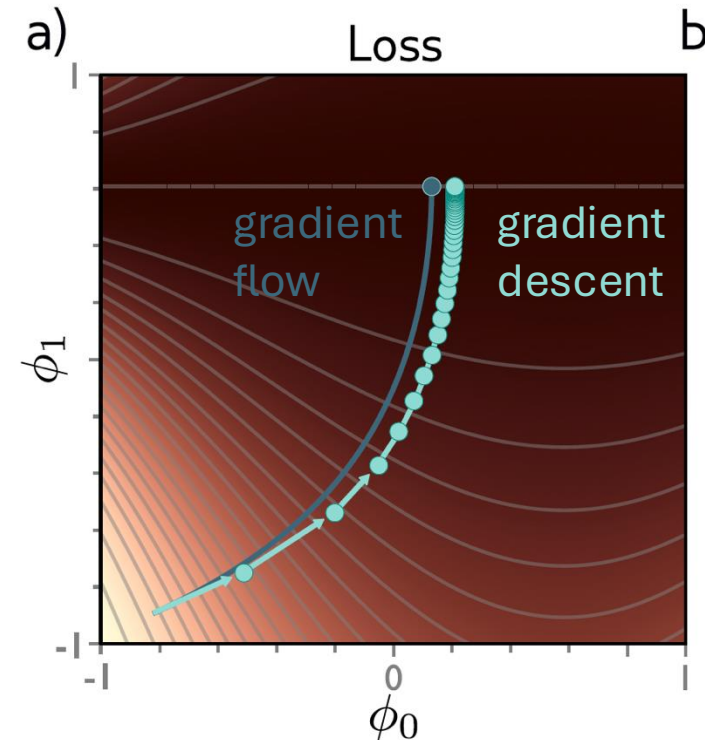
Implicit regularization

$$\phi_{t+1} = \phi_t - \alpha \frac{\partial L[\phi_t]}{\partial \phi}$$

$$\lim_{\alpha \rightarrow 0}$$

$$\frac{d\phi}{dt} = -\frac{\partial L}{\partial \phi}$$

- In the limit, as $\alpha \rightarrow 0$, the gradient descent equation becomes the gradient flow differential equation.
- Doesn't converge to the same place



Implicit regularization

$$\phi_{t+1} = \phi_t - \alpha \frac{\partial L[\phi_t]}{\partial \phi}$$

$$\lim_{\alpha \rightarrow 0}$$

$$\frac{d\phi}{dt} = -\frac{\partial L}{\partial \phi}$$

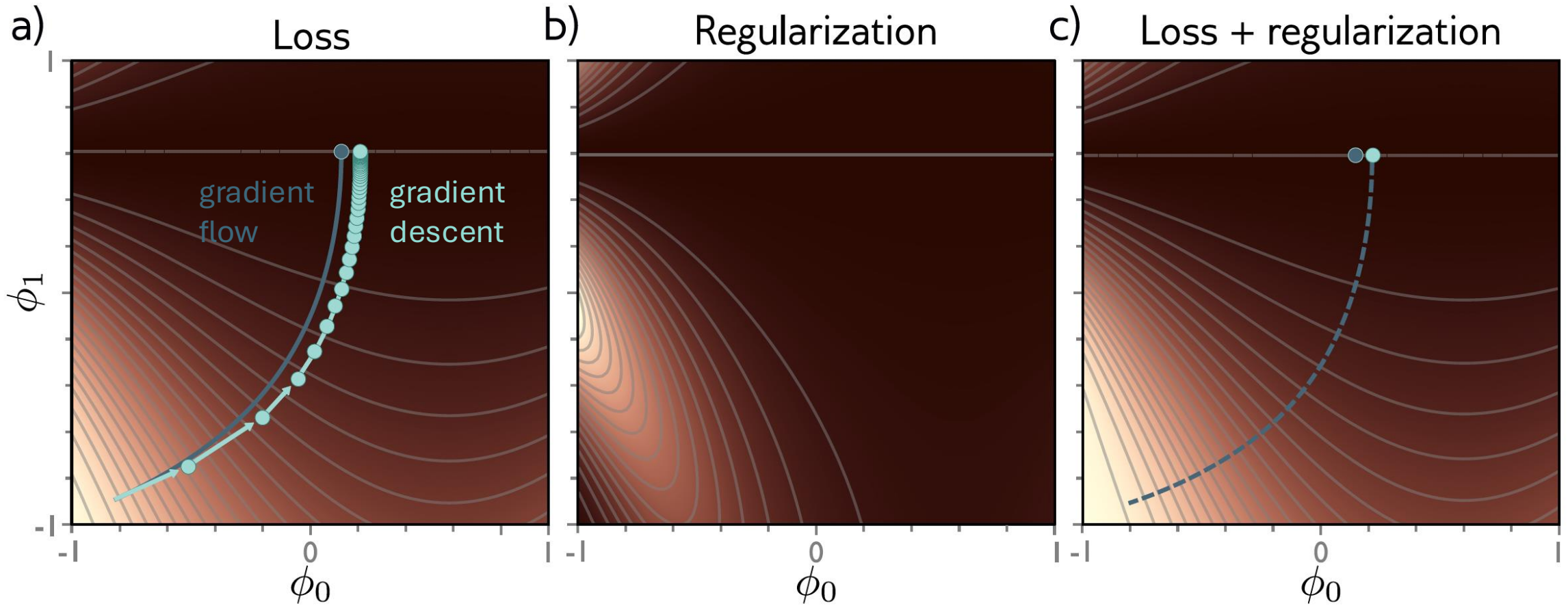
- The implicit regularization can be derived:

$$\tilde{L}_{GD}[\phi] = L[\phi] + \underbrace{\frac{\alpha}{4} \left\| \frac{\partial L}{\partial \phi} \right\|^2}_{\text{Penalty for large gradients.}}$$

See derivation at
end of UDL Ch. 9

Penalty for large
gradients.

Implicit regularization



Gradient descent doesn't converge to same location as (continuous) gradient flow.

Plot of the Implicit regularization ($\sim \|\partial L / \partial \phi\|^2$) to be added to loss

With regularization, continuous descent converges to same place

Implicit regularization of SGD

- Gradient descent disfavors areas where gradients are steep

$$\tilde{L}_{GD}[\phi] = L[\phi] + \frac{\alpha}{4} \left\| \frac{\partial L}{\partial \phi} \right\|^2$$

- SGD likes all batches to have similar gradients

$$\tilde{L}_{SGD}[\phi] = \tilde{L}_{GD}[\phi] + \underbrace{\frac{\alpha}{4B} \sum_{b=1}^B \left\| \frac{\partial L_b}{\partial \phi} - \frac{\partial L}{\partial \phi} \right\|^2}_{\text{batch variance term}}$$

Want the batch variance to be small,
rather than some batches fitting well
and others not well...

Where $L = \frac{1}{I} \sum_{i=1}^I \ell_i[\mathbf{x}_i, y_i]$ and $L_b = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}_b} \ell_i[\mathbf{x}_i, y_i]$.

Implicit regularization of SGD

- Gradient descent disfavors areas where gradients are steep

$$\tilde{L}_{GD}[\phi] = L[\phi] + \frac{\alpha}{4} \left\| \frac{\partial L}{\partial \phi} \right\|^2$$

- SGD likes all batches to have similar gradients

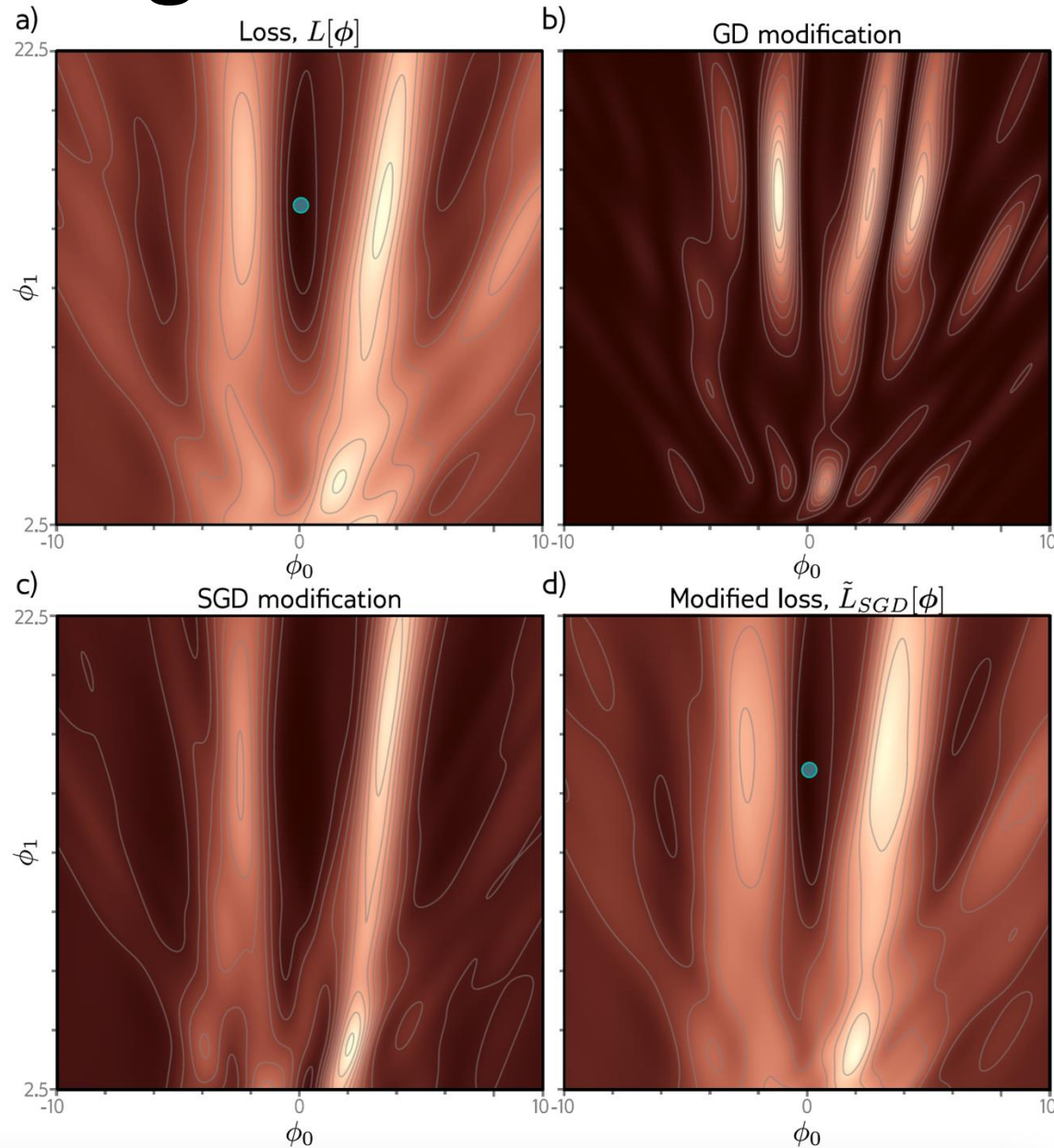
$$\begin{aligned} \tilde{L}_{SGD}[\phi] &= \tilde{L}_{GD}[\phi] + \frac{\alpha}{4B} \sum_{b=1}^B \left\| \frac{\partial L_b}{\partial \phi} - \frac{\partial L}{\partial \phi} \right\|^2 \\ &= L[\phi] + \frac{\alpha}{4} \left\| \frac{\partial L}{\partial \phi} \right\|^2 + \frac{\alpha}{4B} \sum_{b=1}^B \left\| \frac{\partial L_b}{\partial \phi} - \frac{\partial L}{\partial \phi} \right\|^2 \end{aligned}$$

- Depends on learning rate – perhaps why larger learning rates generalize better.
- 

Loss and Regularization Surfaces

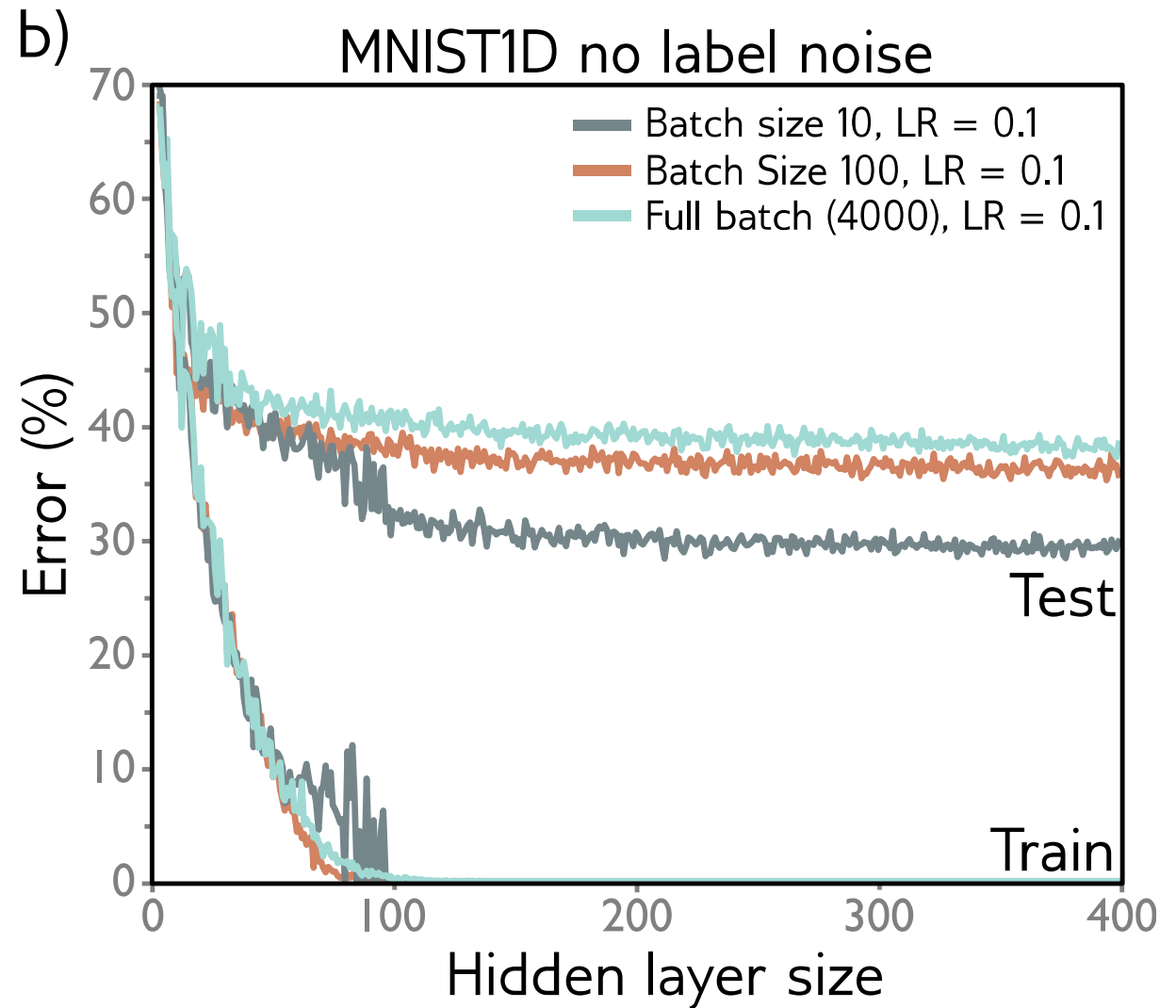
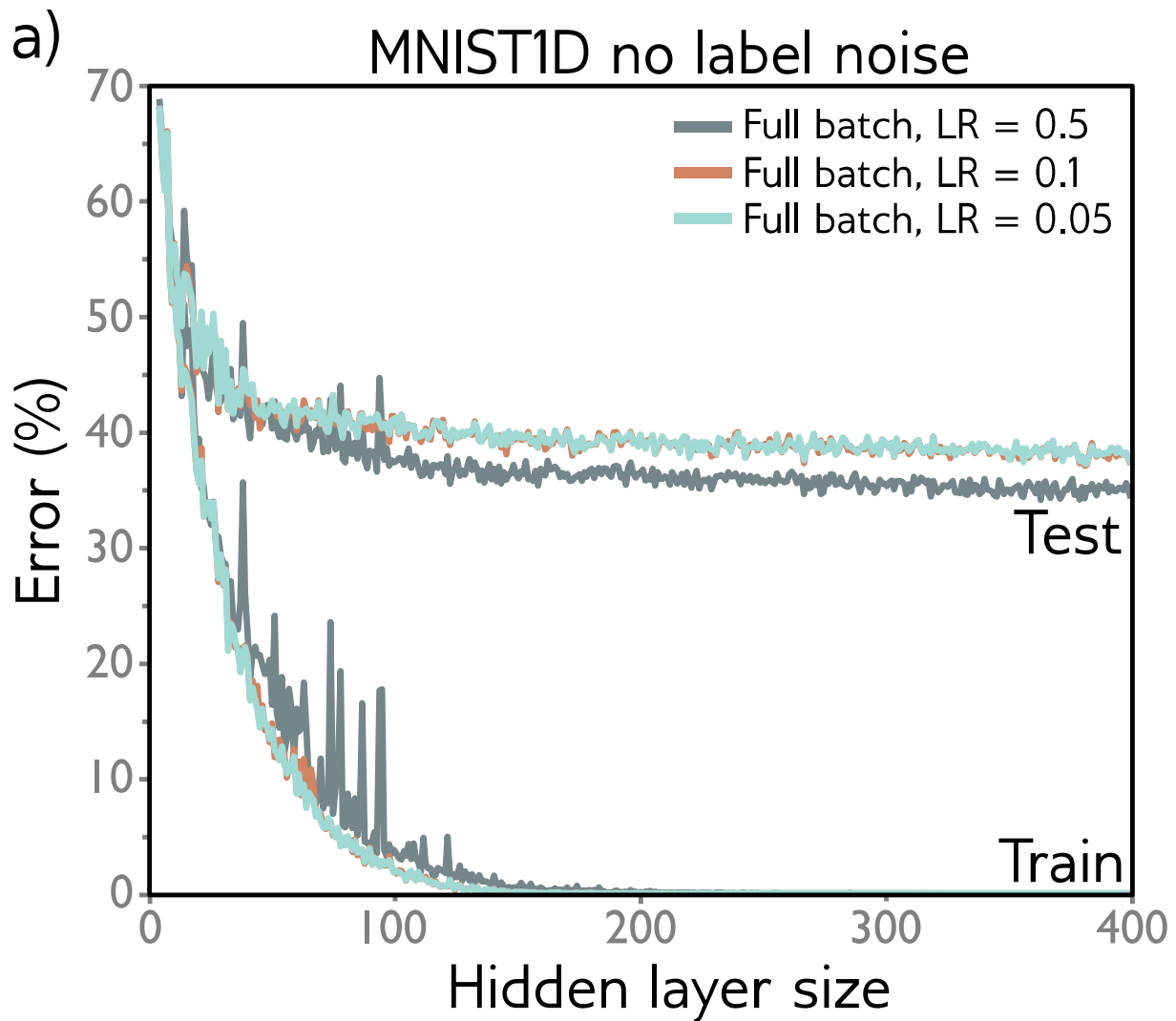
Original Gabor Model
Loss

$$\frac{\alpha}{4B} \sum_{b=1}^B \left\| \frac{\partial L_b}{\partial \phi} - \frac{\partial L}{\partial \phi} \right\|^2$$



$$\frac{\alpha}{4} \left\| \frac{\partial L}{\partial \phi} \right\|^2$$

$$\begin{aligned} \tilde{L}_{SGD}[\phi] &= L[\phi] + \frac{\alpha}{4} \left\| \frac{\partial L}{\partial \phi} \right\|^2 + \frac{\alpha}{4B} \sum_{b=1}^B \left\| \frac{\partial L_b}{\partial \phi} - \frac{\partial L}{\partial \phi} \right\|^2 \end{aligned}$$



Generally, performance is

- best for larger learning rates
- best with smaller batches

Recap: Implicit regularization of GD and SGD

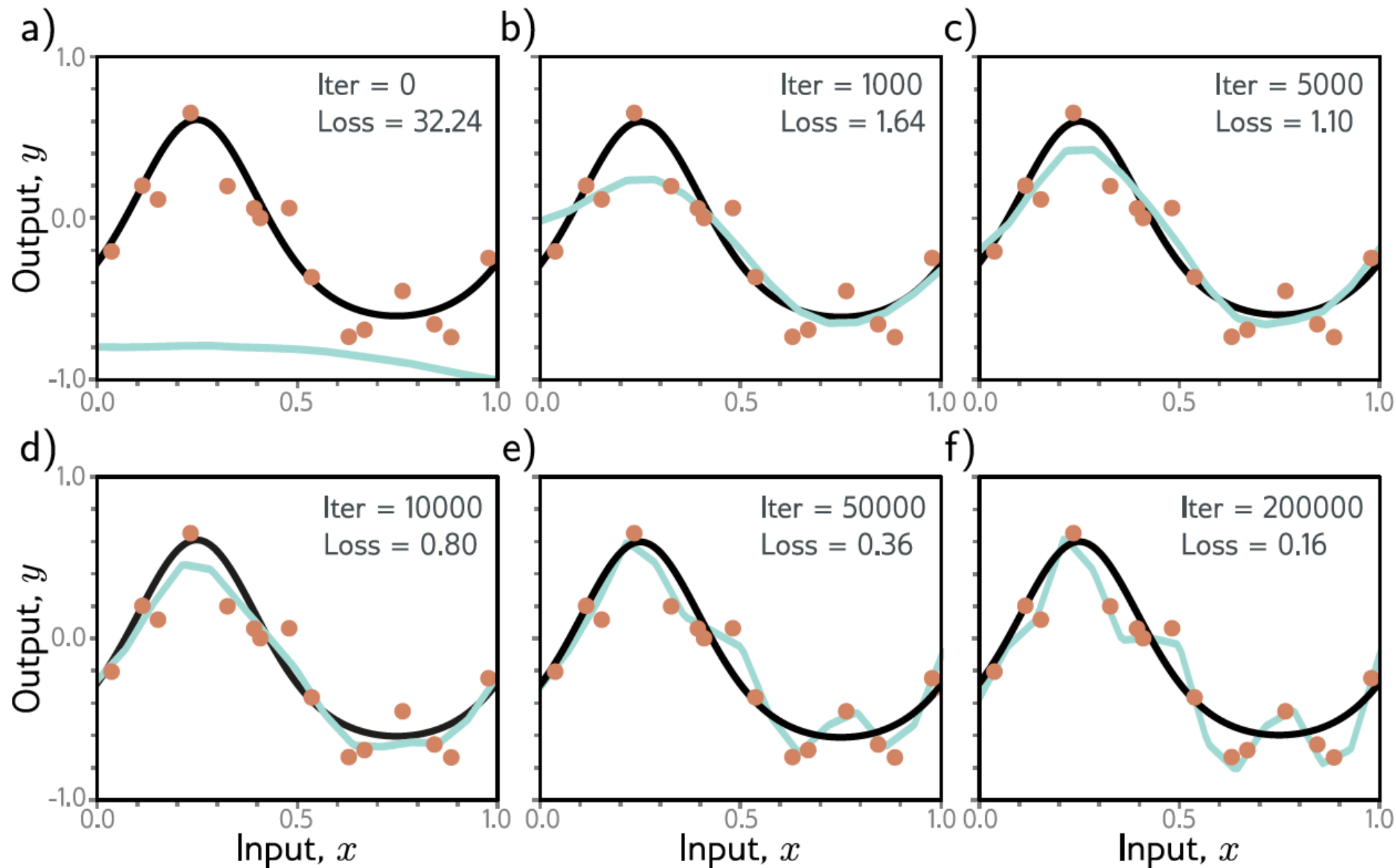
- Larger learning rates may lead to better generalization
- SGD seems to favor places where gradients are stable (all batches agree on slope)
- SGD generalizes better than GD
- Smaller batches in SGD generally perform better than larger ones

Regularization

- Explicit regularization
- Implicit regularization
- Early stopping
- Ensembling
- Dropout
- Adding noise
- Transfer learning, multi-task learning, self-supervised learning
- Data augmentation

Early stopping

- If we stop training early, weights don't have time to overfit to noise
- Weights start small, don't have time to get large
- Reduces effective model complexity
- Known as **early stopping**
- Don't have to re-train with different hyper-parameters – just "checkpoint" regularly and pick the model with lowest validation loss



Simplified shallow network model with 14 linear regions initialized randomly (cyan curve in (a)) and trained with SGD using a batch size of five and a learning rate of 0.05.

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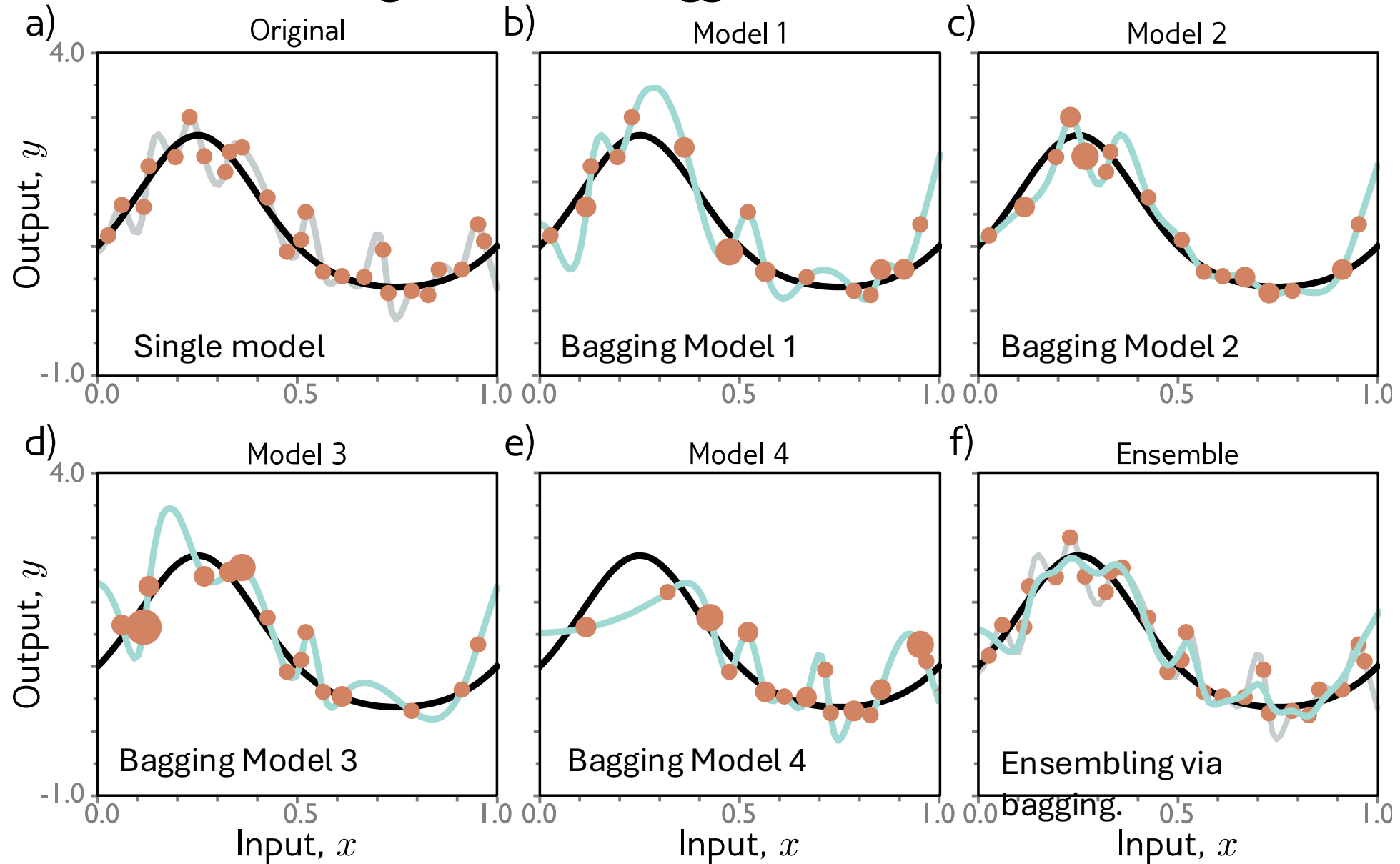
Ensembling

- Combine several models – an **ensemble**
- Combining outputs

	Mean	Median/Frequent (Robust)
Regression	Mean of outputs	Median of outputs
Classification	Mean before softmax	Most frequent predicted class

- Can be simply different initializations or even different models
- Or train with different subsets of the data resampled with replacements – **bootstrap aggregating (bagging)**

Single Model vs Bagged Ensemble



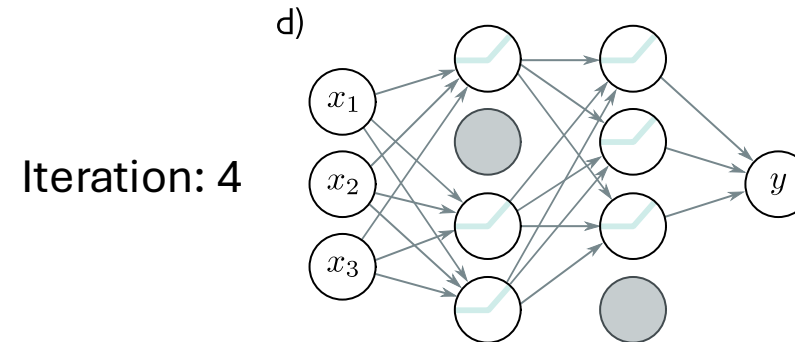
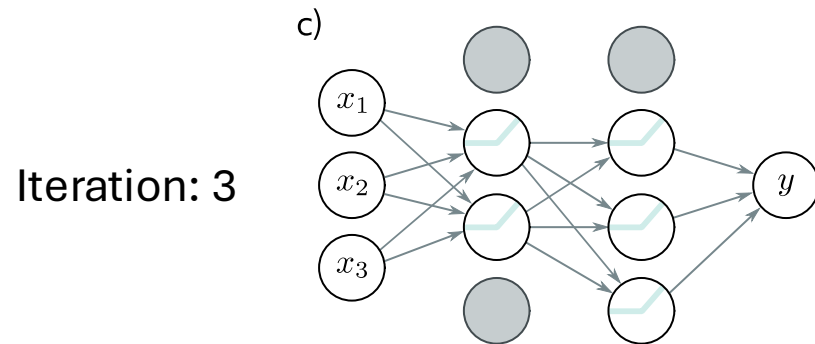
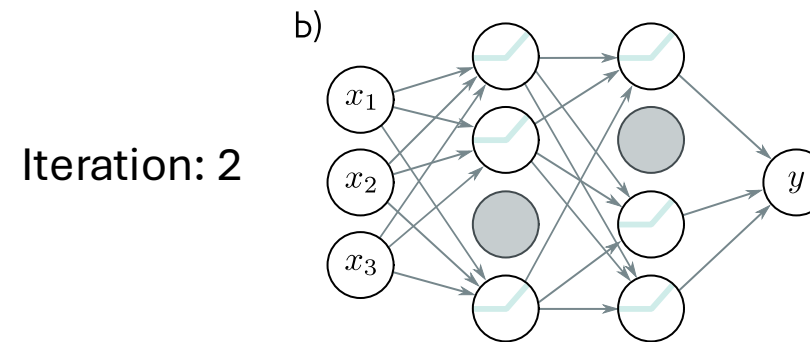
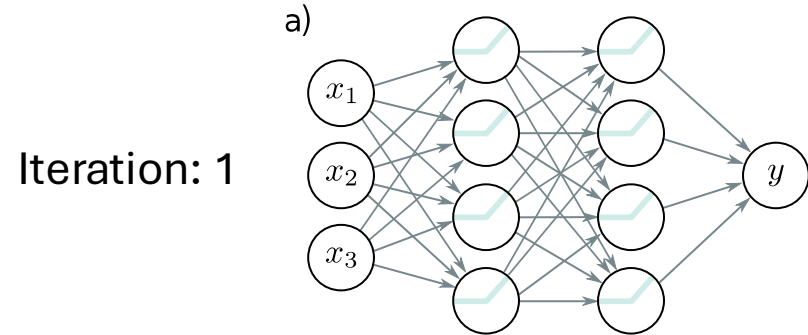
Size of orange point indicates number of times the data point was re-sampled.

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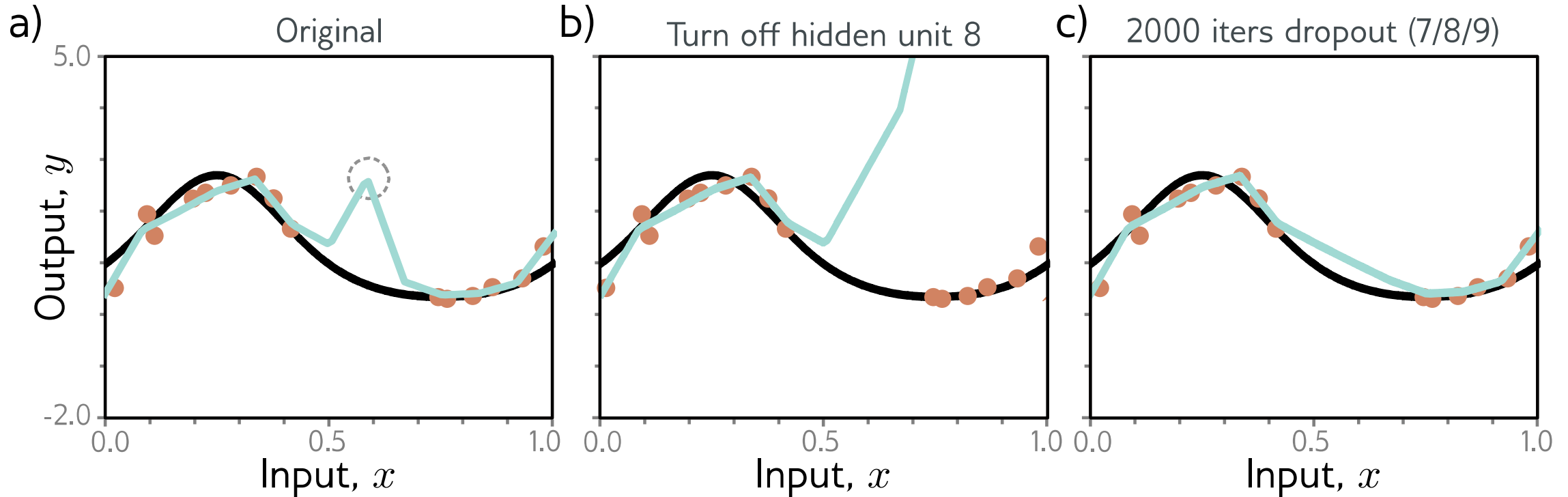
Dropout

Randomly clamp ~50% of hidden units to 0 on each iteration.



- Makes the network less dependent on any given hidden unit.
- At test time, all hidden units are active, which was not the case during training
 - Must rescale using *weight scaling inference rule* – multiple weights by $(1 - \text{dropout probability})$

Dropout



- Prevents situations where subsequent hidden units correct for excessive swings from earlier hidden units
- Can eliminate kinks in function that are far from data and don't contribute to training loss

Monte Carlo Dropout for Inference

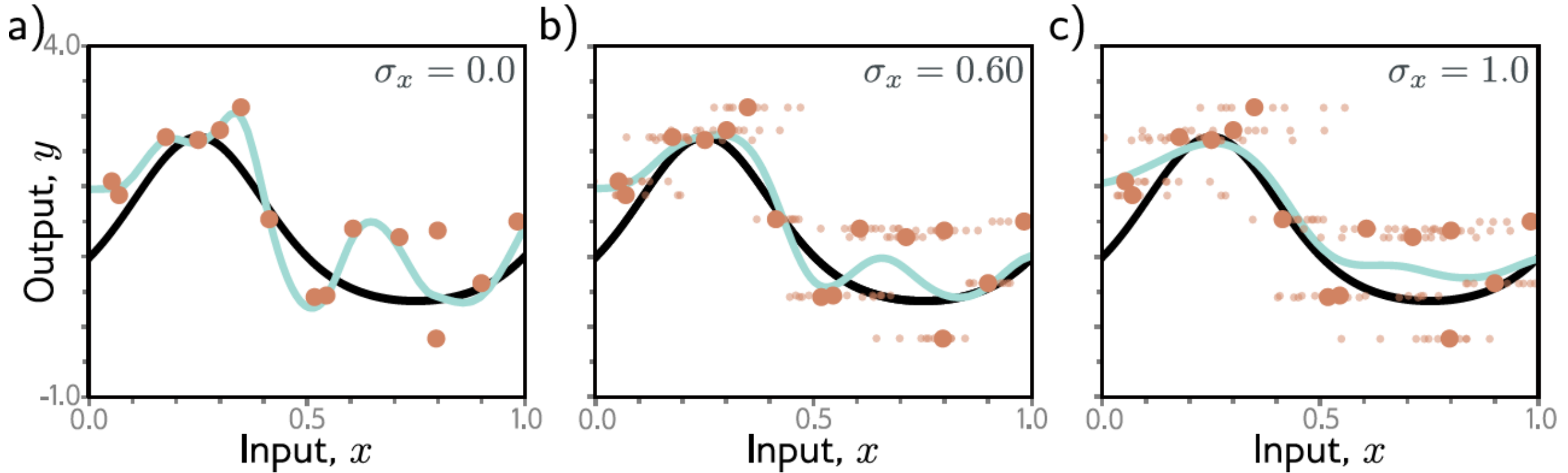
- Run the network multiple times with different random subsets of units clamped to zero (as in training)
- Combine the results using an ensembling method
- This is closely related to ensembling in that every random version of the network is a different model; however, we do not have to train or store multiple networks here.

Regularization

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Adding noise

Adding noise to input with different variances.



- to inputs – induces weight regularization (see Exercise 9.3 in UDL)
- to weights – makes robust to small weight perturbations
- to outputs (labels) – reduces “overconfident” probability for target class

Regularization

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Transfer & Multitask Learning, Augmentation

- Strictly speaking not regularization, but can help improve generalization when dataset sizes are limited

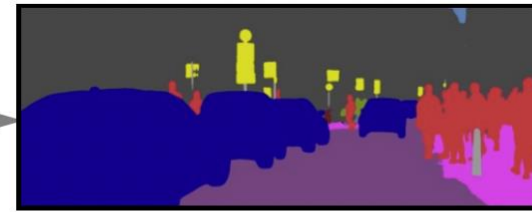
Transfer Learning

(1) Train the model for segmentation



Model

Segmentation
output layer

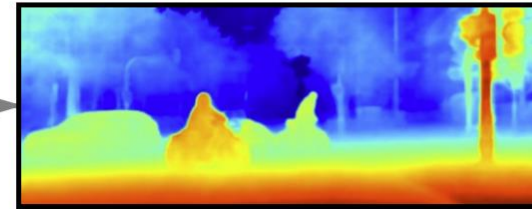


Assume we have lots of
segmentation training
data



Model

Depth
output layer



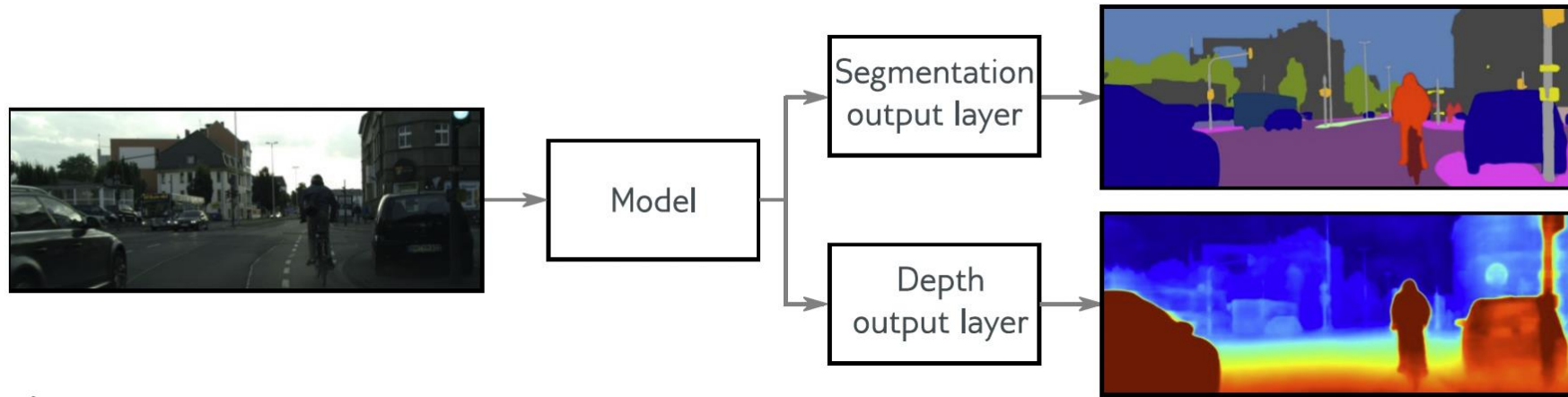
Assume we have
limited depth training
data

(2) Replace the final layers to
match the new task and

(3) Either:

- a) Freeze the rest of the layers
and train the final layers
- b) Fine tune the entire model

Multi-Task Learning

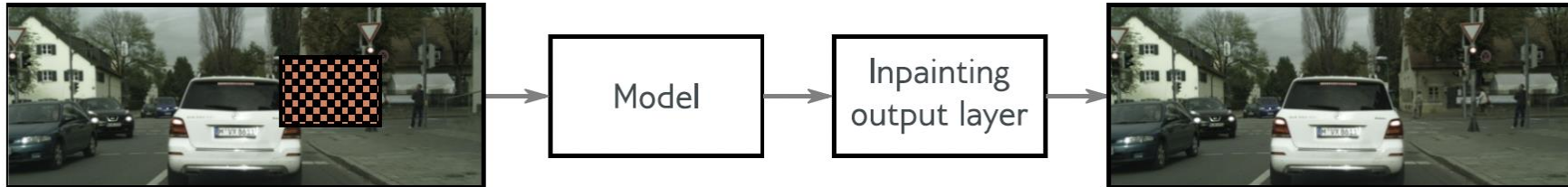


- Train the model for 2 or more tasks simultaneously
 - Weighted combo of loss fncs

$$L_{total} = \alpha \cdot L_{segmentaiton} + \beta \cdot L_{depth}$$

- Less likely to overfit to training data of one task
- Can be harder to get training to converge. Might have to vary the individual task loss weightings, α and β .

Self-Supervised Learning



The animal didn't cross the  because it was too tired.

- Mask out part of the training data
- Train model to try to infer missing data
 - masked data is the target
- ➔ Model learns characteristics of the data
- Then apply transfer learning

Regularization

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Data augmentation

a) Original



b) Flip



c) Rotate and crop



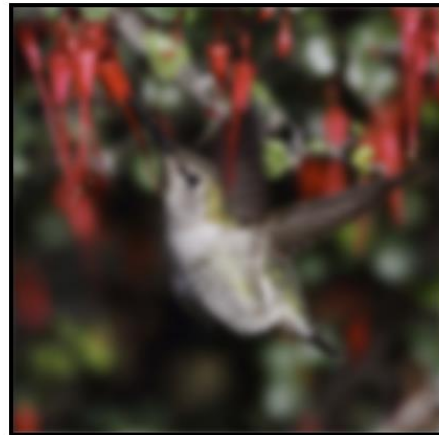
d) Vertical stretch



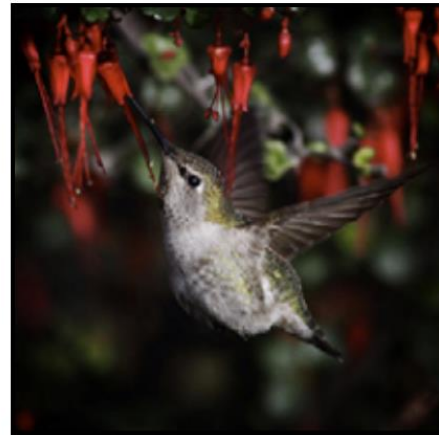
e) Color balance



f) Blur



g) Vignette



h) Pincushion



Image augmentation in PyTorch

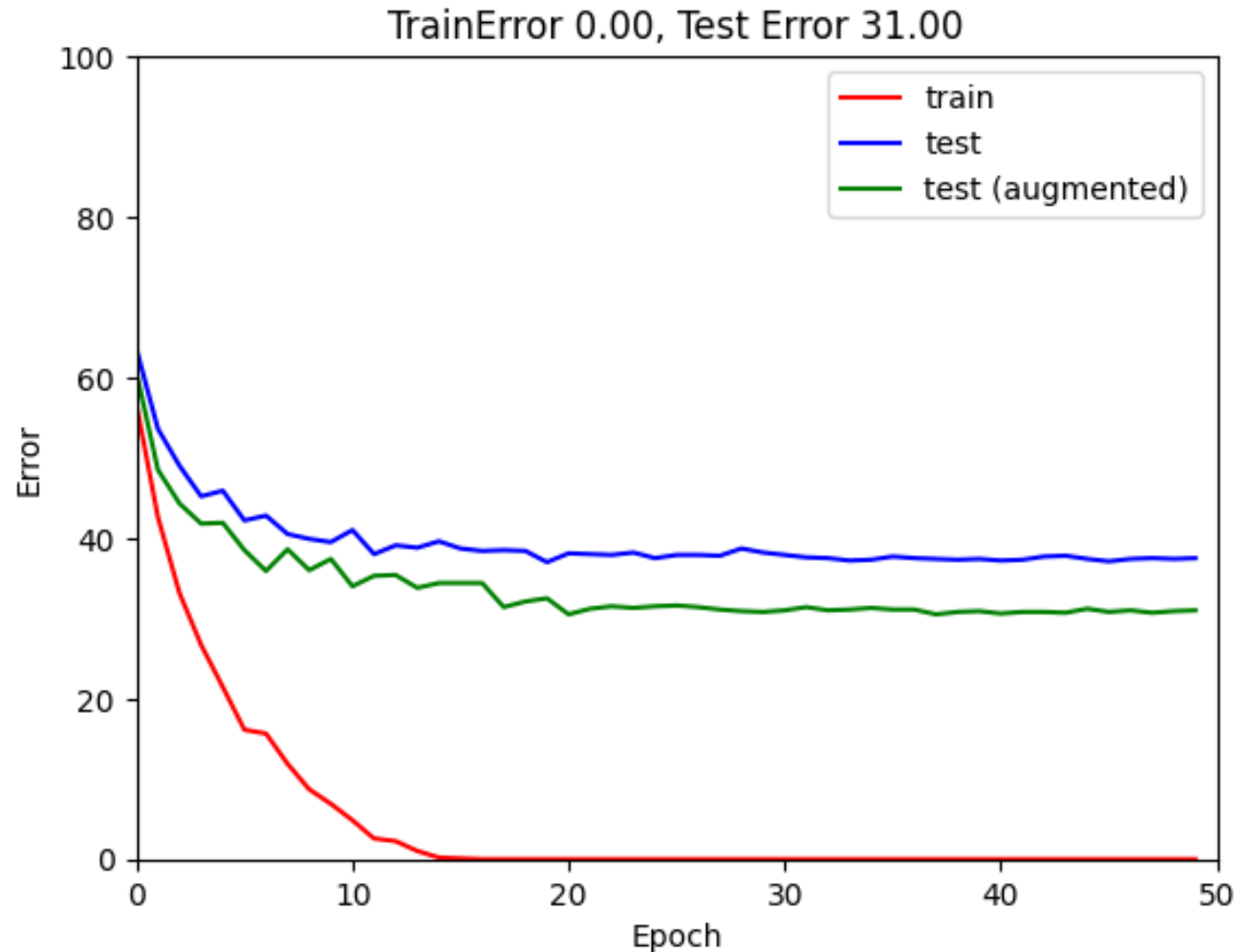
```
import torch
import torchvision.transforms as transforms

# Define augmentation pipeline
transform = transforms.Compose([
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.RandomVerticalFlip(p=0.3),
    transforms.RandomRotation(degrees=30),
    transforms.ColorJitter(brightness=0.5, contrast=0.5, saturation=0.5),
    transforms.RandomAffine(degrees=20, translate=(0.2, 0.2), shear=10),
    transforms.RandomPerspective(distortion_scale=0.5, p=0.5),
    transforms.ToTensor(), # Convert image to tensor
])

# Apply transformations multiple times to visualize augmentation
augmented_image = transform(image)
```

<https://pytorch.org/vision/main/transforms.html>

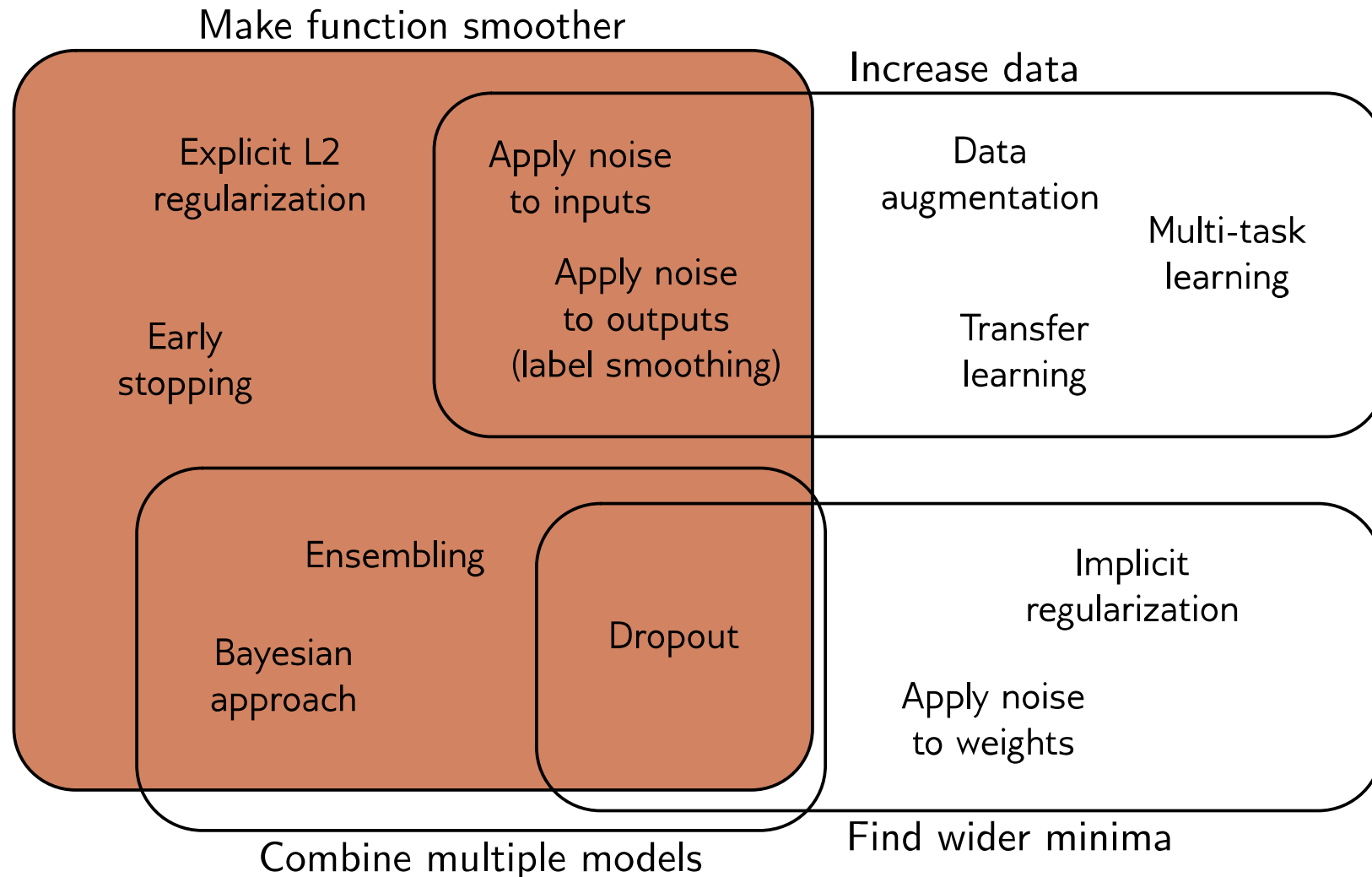
Data Augmentation: MNIST1D



Examples in training set: 4000
Examples in test set: 1000
Length of each example: 40

- Randomly circularly rotate
- randomly scale between 0.8 and 1.2

Regularization overview



Any Questions?

???